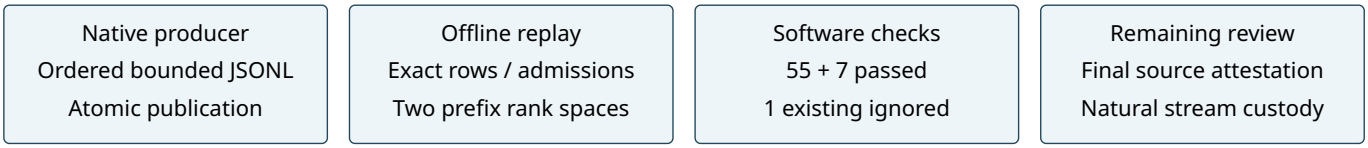
Native pencil event capture and offline replay

Software implementation at parent revision 72e58010d5d7d332b5f42de44c6caf7c1b2f48ef plus the exact source hashes in source-bindings.json. The diagram shows data flow and recorded software checks; it is not a performance plot.



Implemented. Optional native capture writes a target-local header, ordered candidate events, exact submitted sparse relations, actual collector admissions, pending constraints and a terminal summary. Record and serialized-byte limits, sink failure, flush failure and terminal reuse are checked. The CLI publishes through a create-new partial file, successful flush/sync and no-overwrite final link. Source identity is caller-supplied metadata; external custody remains required.

Offline replay. A separately supplied expected header pins workload and configuration. The loader checks domains, column maps, sequence, candidate adjacency, row signs, admissions, pending constraints, residual rank and terminal counters. It reports generated inputs and incorporated inputs separately, with rank(A), rank([A b]) and rank([A b c]). Cap-pending inputs remain separate; exclusion does not claim graph rollback. Scalar candidates and producer geometric claims are explicitly distinguished from independent verification.

Recorded checks. The producer regression passed 55 tests with one existing ignored test. The final consumer regression passed six library test groups and one CLI test. Both retain before/after source bindings. The standalone CLI compiled successfully. Earlier missing-dependency, type-conversion and JSON integer failures remain in the development records. The first root release-library check failed after compiler memory allocation failure under an 8 GiB ceiling; the unchanged-source 16 GiB retry reached its 600-second limit during compilation (exit 124). The full library suite remains unverified; both attempts and source bindings are retained.

Review. A separate reviewer inspected the proposed source contract and identified custody, accounting, memory-bound and pending-row distinctions. The review remains provisional: reviewer availability ended before final source-hash attestation, and root subsequently repaired numeric conversions and cross-crate visibility. Producer tests are not independent source-fidelity review.

Accounting and scope. Capture serialization is outside some measured phase closures, so this diagnostic mode supplies no all-cost timing comparison. The serialized-byte limit is not a process-memory bound. These deterministic software controls do not supply new natural-stream or cohort observations. Signed-graph and Kummer capture adapters, independent final review, runtime qualification and remaining full-cohort work remain separate tasks under their existing limits.

Reproduction and sources. See producer-regression/check.sh, consumer-regression/check.sh and each retained receipt, plus the root-regression directories. The preceding target-stream-reconciliation/SOURCE-MAP.md supplies the inspected construction references and exact row conventions. The consumer README documents expected-header custody and rejection behavior. Prior frozen evidence is unchanged.